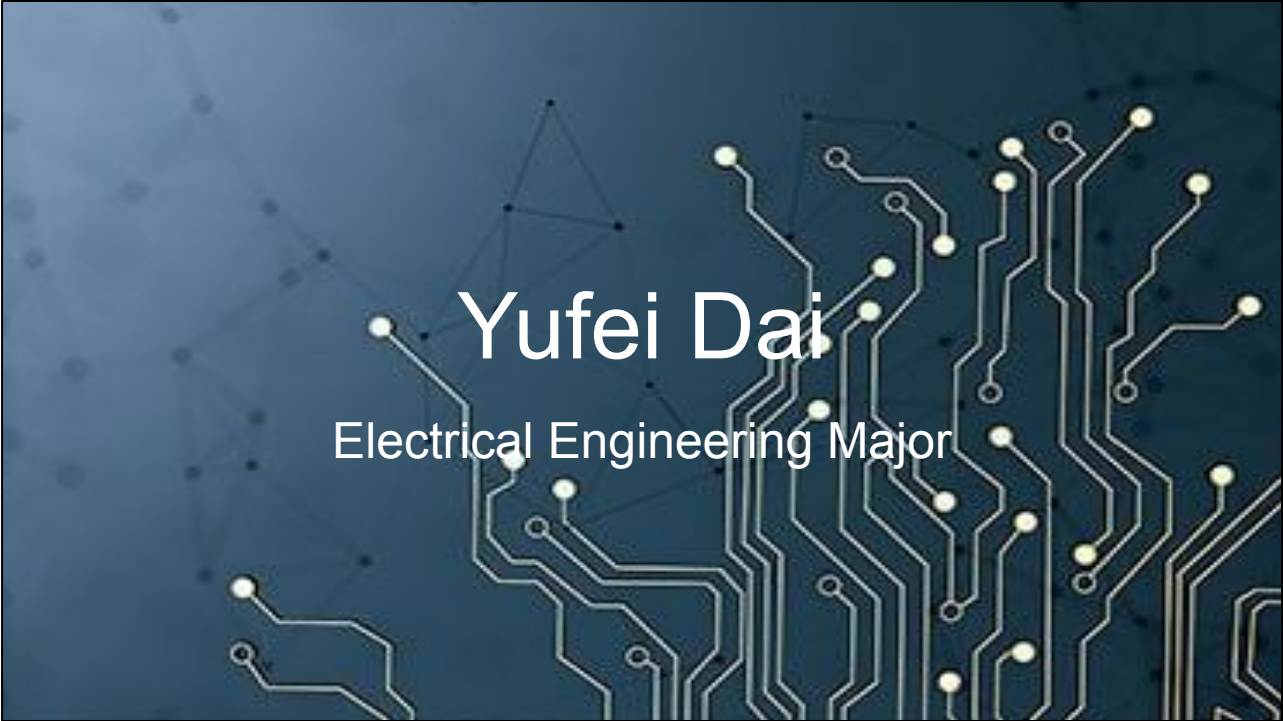




Yufei Dai

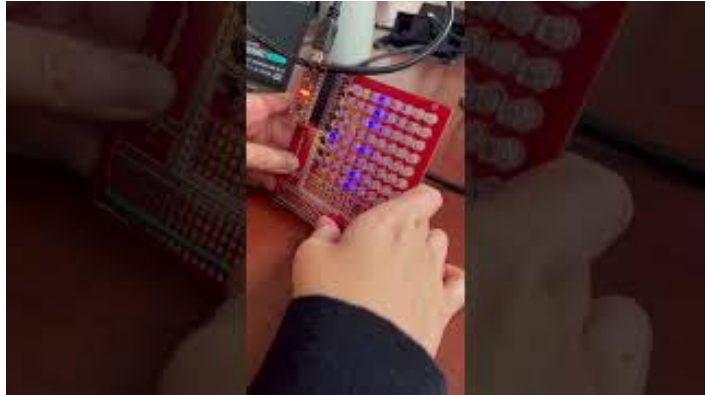
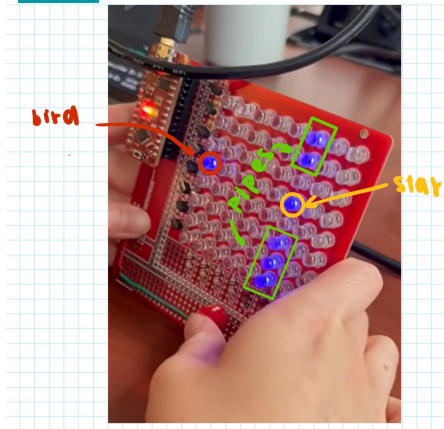
Electrical Engineering Major

Portfolio Walkthrough (15 min)

For the complete code for all programming projects follow this link to github:
<https://github.com/daiyuf25/Yufei-s-Portfolio>

Flappy Bird

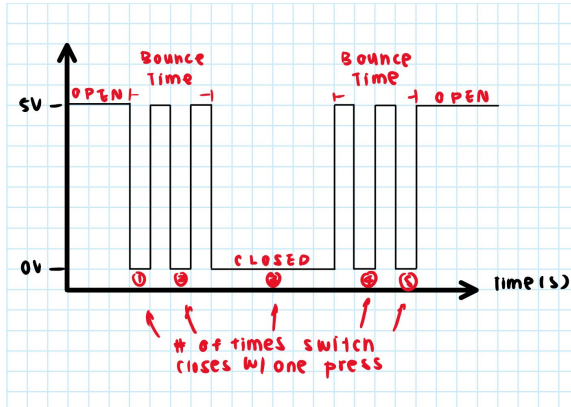
Overview: I recreated flappy bird on an 8x8 LED display board. See here for [demo!](#)



Debouncing the Switch

The push button has a bouncing phenomenon when it gets pressed where it vibrates each time before reaching each steady state.

Signals Arduino Receives from Push Button



The arduino code with register this one button press as 5 different ones.

- The goal is to increase the interval of time between each BUTTON pin reading to avoid having the Arduino mistake the noise as button presses

Debouncing the Switch Using Code

```
const int DEBOUNCE_DELAY = 100;

static int counter = 0;           // count (debounced) button presses
static byte button_state = HIGH;
static byte last_reading = HIGH;
static long last_reading_change = 0;
static char message[50];         // buffer for sprintfs
```

- Initializing variables
- Setting DEBOUNCE_DELAY to 100 allows for noise to be filtered out without missing the actual button presses

Debouncing the Switch Using Code

```
byte reading = digitalRead(BUTTON);

unsigned long now = millis();

// Ignore button_state changes within DEBOUNCE_DELAY milliseconds of the last
// reading change, otherwise accept.

if (now - last_reading_change > DEBOUNCE_DELAY) {

    if (reading == LOW && button_state == HIGH) {

        sprintf(message, "accepted [%d], %ld ms since last\n", ++counter,
            now - last_reading_change);

        Serial.print(message);

        //if the button is pressed move the bird up by one
    }

    button_state = reading;
```

Checks to see if enough time has passed since that last time accepted input from the push button pin

Checks to see if the button was pressed

Debouncing the Switch Using Code

```
// For educational purposes, mention when we ignore a reading change. You
// (probably) don't need this code when you incorporate switch bouncing into
// your cool program.

else if (reading == LOW && last_reading == HIGH) {

    sprintf(message, "ignored, %ld ms since last\n", now - last_reading_change);

    Serial.print(message);

}

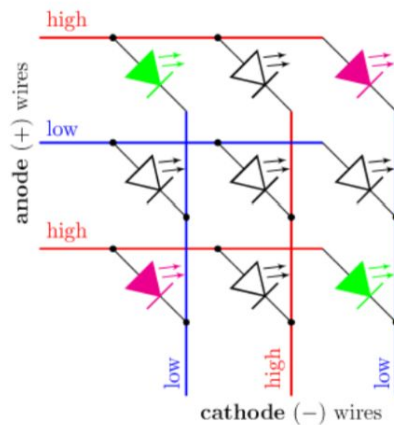
// Prepare for next loop
//rest reading to high if button is released
if (reading != last_reading) last_reading_change = now;
last_reading = reading;
```

Updates the last time the Arduino accepted the signal from the input pin connected to the push button to now



Time Division Multiplexing

Problem with Displaying on LED Lights all at once



- If anode end of the LED is high and the cathode end of the LED is low, then the LED is on
- Turning on the two green diodes will inevitably turn on the two pink diodes which may have not been the intended result
- You can achieve the appearance of just having the two green LED lights on using time division multiplexing

Programing Arduino to Implement Time Division Multiplexing

Pseudocode for display() method

```
for each row (+) wire
  for each column (−) wire
    look up in pattern whether this LED should be on or off
    if LED should be on, activate column (−) wire, else deactivate it
  end for
  activate row (+) wire
  wait a short time
  deactivate row (+) wire
end for
```

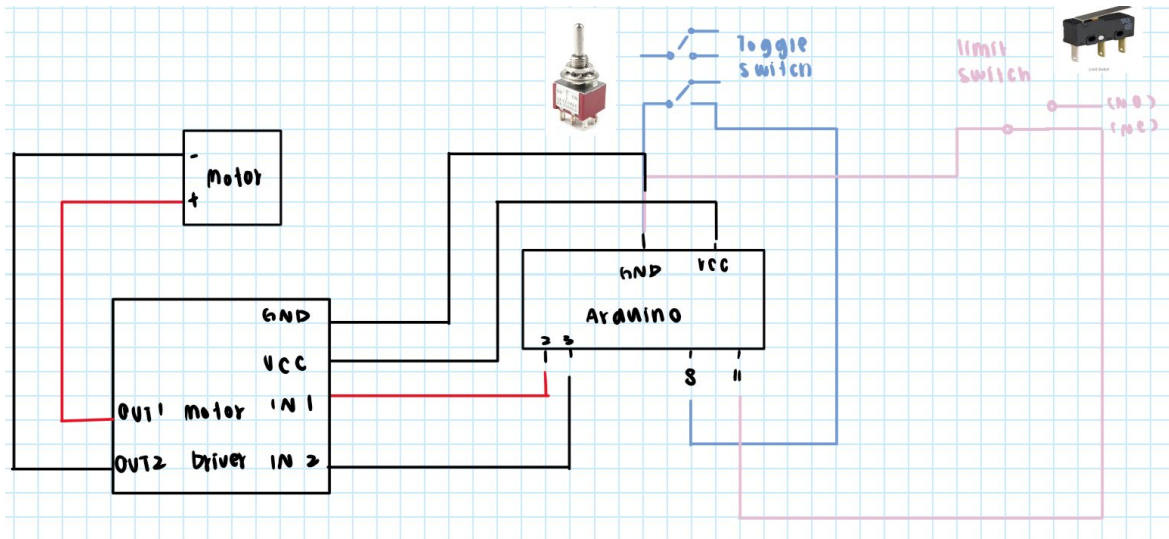
Time Division Multiplexing Method: Only turn on one row of the LED display at a time and rapidly rotate which row is on so that only the LED lights meant to turn on appear like they are on at the same time.

Useless Box

Overview: The useless box is a toy where you flip a switch on the box and a finger in the box flips the switch back. I changed the Arduino code to give my useless box a personality. This [video](#) is a demo of the the useless box I made.



Electrical Schematic



High Level Overview

- TOGGLE switch and LIMIT switch send signals to the Arduino
- The Arduino uses the signals from the switches to determine signals to send out from pins 2 and 3

Code Overview

Setting up the Arduino Pins

```
void setup() {  
  Serial.begin(115200);  
  Serial.println("Setup complete");  
  
  // TO DO:  
  // initialize input and output pins  
  pinMode(TOGGLE, INPUT_PULLUP);  
  pinMode(LIMIT, INPUT_PULLUP);  
  pinMode(two, OUTPUT);  
  pinMode(three, OUTPUT);  
  
  // initialize your initial states  
  state = STOP;  
}
```

Loop() Structure

```
switch (state) {  
  case STOP:  
    // code  
    break;  
  
  case FORWARD:  
    // code  
    break;  
  
  default:  
    // code  
    break;  
}
```

setup()

- Set the Arduino pins as input or output
- Sets initial state to STOP

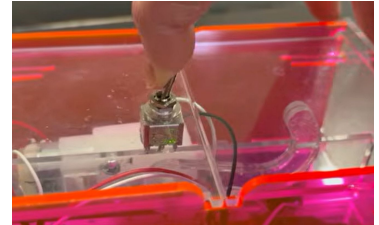
loop()

- The code in inside of the loop is a switch case statement

Stop Case

```
case STOP: // case for STOP
  // TO DO: Set your outputs for the motor to stop
  digitalWrite(two, LOW);
  digitalWrite(three, LOW);

  // TO DO: Decide whether to change states and update the state variable accordingly
  if (digitalRead(TOGGLE) == LOW){
    state = FORWARD;
  }
  break;
```



STOP Case:

- sets both output pins to be LOW
- STOP -> FORWARD: If the user flips the switch, TOGGLE pin will read LOW which switches the state to FORWARD

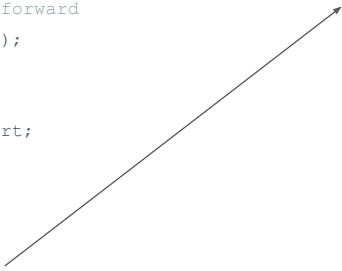
Forward/Reverse Case

```
case FORWARD: // case for forward
    Serial.println("FORWARD");

    int now = millis();
    int interval = now - start;
    Serial.print("start ");
    Serial.println(start);
    Serial.print("now ");
    Serial.println(now);

    if (0 <= interval && interval <
        INTERVAL_SHY) {
        Serial.println("SHY_CASE");

        //start moving slowly
        for (int i = 0; i <= 15; i++) {
            digitalWrite(three, LOW);
            digitalWrite(two, HIGH);
            delay(10);
            digitalWrite(three, LOW);
            digitalWrite(two, LOW);
            delay(30);
            Serial.println("forward");
        }
    }
```



FORWARD Case:

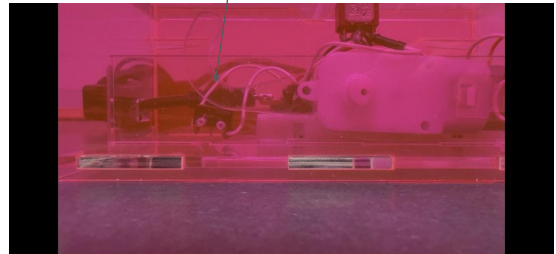
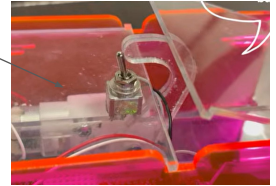
- 1) The signals sent through the output pins depended on how long the program has been running
- 2) Let's look at the shy interval.
 - First the the finger moves counterclockwise at a slow speed.

Forward/Reverse Case

```
//continue moving
for (int i = 0; i <= 15; i++)
    digitalWrite(three, LOW);
    digitalWrite(two, HIGH);
    delay(20);
    digitalWrite(three, LOW);
    digitalWrite(two, LOW);
    delay(10);
    Serial.println("forward");
}

digitalWrite(three, LOW);
digitalWrite(two, HIGH);
Serial.println("forward");
delay(500);

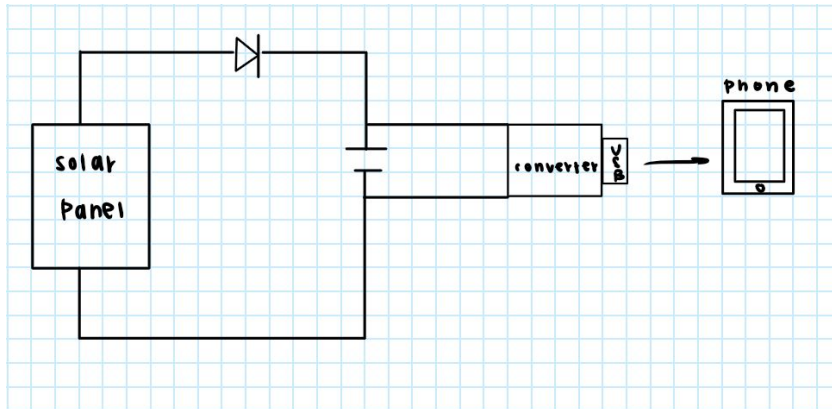
if (digitalRead(TOGGLE) == HIGH){
    digitalWrite(two, LOW);
    digitalWrite(three, HIGH);
    delay(1000);
    if (digitalRead(LIMIT) == HIGH){
        Serial.println("stop");
        state = STOP;
    }
}
```



- The finger then pauses before continuing forward as a quicker pace
- When the finger flips back the toggle switch, TOGGLE reads HIGH which signals the code to tell the finger to reverse
- When the finger hits the limit switch, the LIMIT will read high which will switch the state to STOP

Solar Panel Phone Charger

Schematic:



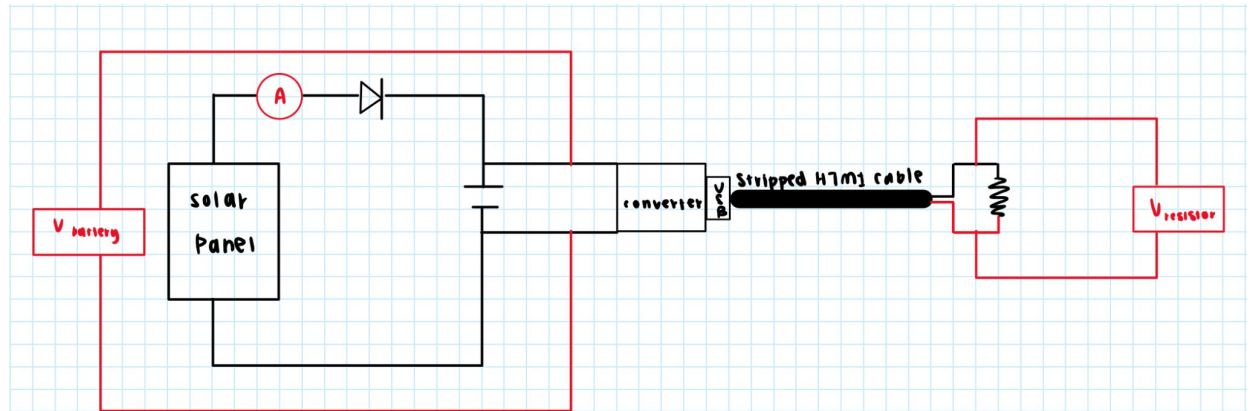
Picture:



The solar panel is used to charge a 3.7V battery. The converter increases the battery voltage to 5V for phone charging. The phone only accepts voltages around 5V. A diode is placed between the solar panel and the battery to prevent current from flowing out of the battery and back into the solar panel when it is dark. This is called reverse current which can damage the solar panel.

On the right is a picture of the device in action.

Finding True Converter Efficiency



This is the set up showing where the voltmeters and ammeters were placed in the circuit in order to gather the data needed to determine the efficiency of the converter.

Method

	Battery			Resistor			
	Voltage	Current	Power	Voltage	Resistance	Power	Efficiency
82Ω	3.9	0.12	0.468	5.1	81	0.321	0.686
27Ω	3.81	0.28	1.0668	5.03	26	0.973	0.912
13.5Ω	3.05	0.7	2.135	4.86	13	1.814	0.851

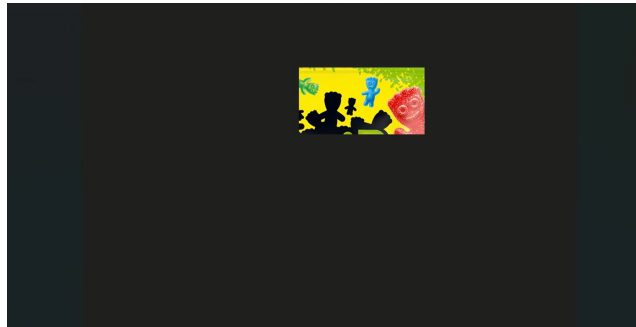
$$P_{Battery} = i_{Battery} * v_{Battery} = P_{in} \quad P_{Resistor} = \frac{v_{Resistor}^2}{R} = P_{out}$$

$$Efficiency = \frac{P_{out}}{P_{in}}$$

The converter board's website says that the efficiency is 92%. The highest efficiency, 91.2%, is close to the advertised efficiency.

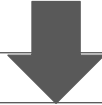
Guess the Picture with Python

Overview: This code reveals one small section of a picture per round to a user who tries to guess what is depicted in the picture. The code makes an API call to ChatGPT to verify the user's answers. This helps account for the variability of user's responses that may not exactly match the answer given by the coder.



Revisions

```
def get_answer_validation(self, guess):
    chat_completion = openai.chat.completions.create(
        messages=[{
            "role": "user",
            "content": "My friends are playing guess the image. The correct answer is " + self._correctanswer + "."
            " Would " + str(guess) + " be a valid answer too? Please be lenient and please just respond yes or no."
        }],
        model="gpt-4",
    )
```



```
def get_answer_validation(self, guess):
    chat_completion = openai.chat.completions.create(
        messages=[{
            "role": "user",
            "content": "My friends are playing guess the image. The correct answer is " + self._correctanswer + ". "
            "Would " + str(guess) + " be a valid answer too? Please be lenient but don't allow answers that are not specific enough. "
            "Please just respond yes or no."
        }],
        model="gpt-6-astra",
    )
```

I upgrade the version of ChatGPT from GPT-4 to GPT-6 Astra and I also revised the prompt. The combination significantly improved the game's ability to evaluate user's answers.

Answer Validation w/ GPT-6 Astra



Answer Validation Results

Agreed with ChatGPT	Disagreed with ChatGPT
✓ Sweet gummy candy with sour coating shaped like a kid	✗ gummy candy
✓ suor patch kdis	
✗ candy	
✗ colorful stuff	

Game Demos

[Demo 1](#)



[Demo 2](#)



Using the API Call

- Code in Guess the Picture.py (main program)
- ImageGuessingAssistant.py

```
assistant = ImageGuessingAssistant(pictures[name_image_to_guess])
```

```
def __init__(self, answer):  
    self._correctanswer = answer  
    self._hints = []
```

Create an image object by
passing in the answer for
the current picture in play

```
while len(puzzle_pieces_list) != 1 and assistant.get_answer_validation(guess) == False:
```

Calling the method `get_answer_validation` in `ImageGuessingAssistant.py`

- The class called `ImageGuessingAssistant.py` has a constructor which creates an `ImageGuessingAssistant` object
- The code calls methods on the `ImageGuessingAssistant` object to get responses from ChatGPT

Using the API Call

```
def get_answer_validation(self, guess):
    chat_completion = openai.chat.completions.create(
        messages=[{
            "role": "user",
            "content": "My friends are playing guess the image. The correct answer is " + self._correctanswer + ". "
            "Would " + str(guess) + " be a valid answer too? Please be lenient but don't allow answers that are not specific enough. "
            "Please just respond yes or no."
        }],
        model="gpt-6-astra",
    )
    response_str = chat_completion.choices[0].message.content
    if response_str.strip(".").lower() == "yes":
        return True
    return False
```

Correct answer

User guess

Return whether or not the user's guess is valid

Thank You!